

12. Camera Driver

12. Camera Driver

- 12.1 Experiment Objective
- 12.2 Experiment Preparation
- 12.3 Experiment Procedure
- 12.4 Experiment Summary

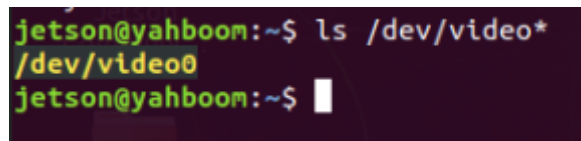
12.1 Experiment Objective

This course focuses on using the main board's camera to capture video and display it on a JupyterLab widget.

12.2 Experiment Preparation

Check the device number of the USB camera. As shown in the figure below, the device number is `/dev/video0`.

```
ls /dev/video*
```



```
jetson@yahboom:~$ ls /dev/video*  
/dev/video0  
jetson@yahboom:~$
```

12.3 Experiment Procedure

Open the JupyterLab client and navigate to the code path:

```
muto/Samples/Control/12.camera.ipynb
```

Import the OpenCV library and JupyterLab display widgets.

```
import cv2  
import ipywidgets.widgets as widgets
```

Turn on the camera. The default device is `/dev/video0`, with a display resolution of 640x480 and a frame rate of 30 fps. If the camera's device number in the system is not `/dev/video0`, please modify the configuration information according to the actual device number.

```
image_widget = widgets.Image(format='jpeg', width=640, height=480)  
image = cv2.VideoCapture(0)  
image.set(3, 640)  
image.set(4, 480)  
image.set(5, 30)
```

Read one frame from the camera. `ret=True` on success, `ret=False` on failure. `frame` represents the currently read image. Calculate the FPS and display the image on the widget.

```
while g_camera.isopened():
    ret, frame = g_camera.read()
    if not ret:
        print("Camera Read Fail")
        break

    m_fps = m_fps + 1
    fps = m_fps / (time.time() - t_start)
    if (time.time() - t_start) >= 2:
        m_fps = fps
        t_start = time.time() - 1

    text="FPS:" + str(int(fps))
    cv2.putText(frame, text, (10, 20), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0,
255, 255), 1)
    image_widget.value = bgr8_to_jpeg(frame)
```



When not in use, please release the camera to avoid errors from other routines accessing it simultaneously. Select the cell that displays the camera feed and click the stop button to automatically stop and call the camera release program.



```
g_camera.release()
```

12.4 Experiment Summary

This experiment uses a JupyterLab widget to display the camera feed. If the default device `/dev/video0` cannot be driven, please check the current system's device number or try re-plugging the camera cable.

